Orbital Advection Across Azimuthal Level Jumps in the AthenaK Shearing Box

Validation Document

`athenak-multigrid` tree, branch `fargo-amr`
parent commit `d3edb948` (branch `dust-multigrid`)

October 28, 2026

Abstract

Orbital advection (FARGO) in the AthenaK shearing box now accepts x_2 neighbors one level coarser or finer, so refinement may cover only part of the azimuth (method: `fargo_amr_implementation.pdf`). This document records the tests. Every annular and uniform configuration of the earlier test suite, hydro and MHD, is bit-identical to the parent code. With the orbital-advection flux correction the remap conserves mass to round-off across static and moving level jumps ($\sim 5 \times 10^{-14}$ over hundreds of steps; 10^{-7}–10^{-6} without it). Serial and 3-, 4- and 7-rank runs are bit-identical, and the same tests run on an H200 GPU. A spatially uniform epicycle stays exactly uniform on a refined patch. Vortical shearing waves on a static patch and under a moving box lie between the ring and coarse-grid results, with no visible artifact at the level jumps. Swing amplification with multigrid self-gravity on a patch moves the peak 0.60% toward linear theory, and a gravity-off control shows that gravity adds no deviation of its own. A timestep limit lets a run proceed where the shift would otherwise exceed the neighbor's rows. For MHD (branch `fargo-amr-mhd`) orbital advection now runs on refined meshes, rings and partial-x_2 patches alike: the remap's EMFs are corrected at level jumps, which keeps the two copies of every shared face equal to round-off (10^{-2} apart without it), and the refined-mesh field error of the magnetized swing (13–22% without orbital advection) is gone: refined runs follow their uniform twins to 1–3% and linear theory better than the uniform runs. Pushed into collapse ($4\pi G\rho_0 = 6$) with a Jeans-length criterion (16 cells, three levels), patches and rings give the same collapse — the same site, the same time, winding waves within 0.012 of each other where refinement itself moves them by 0.13 — with 8.6 times fewer MeshBlock-cycles for patches; both collapse earlier than a uniform 256^2 run, because between the crests the criterion leaves the box at the 128^2 root. Magnetized ($\beta = 8$), the refined runs collapse at the centre, 1.2 time units before a uniform 256^2 run, which collapses on the shear-periodic x_1 boundary, where the mesh policy forbids refinement. AMR regrids with two or more levels leave the two copies of shared faces 10^{-7}–10^{-3} apart, with or without orbital advection; like the $\nabla\cdot B$ jumps at derefinement (10^{-13}–10^{-12} per event next to fine x_2 neighbours), this is a pre-existing AthenaK defect, documented here. The production test — SC14 clump-following AMR with and without rings — is still to be run.

Contents

1 Test ladder **3**

2 Regression: nothing changes where nothing jumps **3**

3 The epicycle on a patch **3**

4 Exact conservation **4**

5 MPI **5**

6 Vortical shearing wave through static and moving level jumps **5**

7 Swing amplification with multigrid gravity on a patch **5**

8 The timestep limit **6**

9 MHD with orbital advection on refined meshes **7**
9.1 Regression . . . 7
9.2 The refined-mesh field error disappears with orbital advection . . . 7
9.3 Shared faces stay consistent . . . 8
9.4 Magnetized swing amplification with gravity . . . 9
9.5 Across x_2 level jumps . . . 12

10 GPU **13**

11 Nonlinear swing to collapse **13**

12 Not yet validated **17**

13 Reproduction **18**

A Linear theory of the swing tests **19**
A.1 Background . . . 19
A.2 Linearized equations . . . 19
A.3 Shearing waves . . . 20
A.4 The discrete Laplacian . . . 20
A.5 Checks . . . 21
A.6 Projections and the comparison . . . 21

1 Test ladder

All runs are on a Mac (Apple Silicon, serial build and Open MPI build), October 28, 2026, except those of §10 and §11 (one H200 GPU). Run directories are in `validation/run/fargo_amr/`; its `README.md` lists the command for each one. "Patch" means a refined region that covers only part of x_2, so it has x_2 level jumps; "ring" means the old annular configuration.

Test	What it probes	Result
Regression (§2)	same-level path unchanged	bit-identical
Epicycle on a patch (§3)	buffers, indices, prolongation	exact
Conservation (§4)	flux correction, static and AMR	5×10^{-14}
MPI (§5)	per-sub-face messages and tags	bit-identical
Vortical shearing wave (§6)	structure through level jumps	between ring and coarse
Swing with gravity (§7)	multigrid at x_2 level jumps	+0.60% toward theory
Timestep limit (§8)	large shifts	runs, conserves
MHD on rings (§9)	orbital EMF correction, field error	faces to 10^{-15}; 1–3%
MHD across x_2 jumps (§9.5)	B_1, B_3 rows across levels	faces to 10^{-15}; 1–3%
Swing to collapse (§11)	Jeans-criterion AMR, patches vs rings	same collapse, 8.6× fewer blocks
SC14 production (§12)	cost	not yet run

2 Regression: nothing changes where nothing jumps

On a same-level face the new code performs the same arithmetic as before, only with a different buffer layout and depth. Each input below was run with a build of the parent commit and with the new build, and the binary dumps (the data after the parameter header, which differs by the added input line) and history files compared byte by byte. All are **bit-identical**:

Input (`inputs/shearing_box/`)	Configuration
`epicycle_smr_ring`	level-1 ring, 1 orbit
`epicycle_amr_ring`, `epicycle_amr_ring2`	moving ring, 1 and 2 levels, incl. block create/destroy sequence
`hydro_incompress_shwave_smr_ring{,2}`	ring, 1 and 2 levels
`hydro_incompress_shwave_amr_ring`	moving ring
`epicycle`, `hydro_incompress_shwave`, `hydro_compress_shwave`	uniform mesh (VTK dumps for the shwaves)
`mhd_compress_shwave`	uniform mesh, MHD with orbital advection

The AMR ring inputs were run with `oa_annular = true`, which they need for the ring-wise flag synchronization (now off by default). The new orbital-advection timestep limit does not bind in any of them.

3 The epicycle on a patch

`epicycle_smr_ypatch.athinput`: $20 \times 20 \times 1$ box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks; the level-1 region covers $|x|, |y| < 5$, two of the four block rows in y, so four x_2 faces are level jumps. The `ipert=1` epicycle is spatially uniform at all times, so any error in buffer layout, indexing or prolongation appears as $O(1)$ structure. After one orbit (365 steps):

- density, all velocity components and the internal energy are uniform to the last digit of the single-precision output in every one of the 44 blocks;
- every history column matches the ring run (`epicycle_smr_ring`), and the mass does not change;
- on 4 MPI ranks every dump is bit-identical to the serial run.

4 Exact conservation

A uniform field cannot show a conservation error, and the vortical shwave of §6 has amplitude 10^{-4} and constant density. The conservation test is therefore the compressible shwave (`hydro_compress_shwave_smr`, isothermal, HLLE, PLM) at amplitude 0.1 on a 4×4 box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks, with a level-1 patch over $|x|, |y| < 1$. The x_1 boundaries are plain *periodic*: the shear-periodic remap at the x_1 faces is itself conservative only to $\sim 10^{-8}$ and would hide the effect. Only the hydro update and the orbital remap then move mass. The moving-box variant uses the AMR criterion of the implementation document with $x_c = 0.9 \sin t$, $y_c = 0.8 \sin 2t + t$ and half-widths 0.3.

maximum $\lvert M(t) - M(0)\rvert / M(0)$	flux correction on	off
static patch, $t = 0$–2 (453 steps)	5.2×10^{-14}	1.8×10^{-7}
moving box (AMR), $t = 0$–4 (899 steps; 98 blocks created, 84 destroyed)	5.7×10^{-14}	1.0×10^{-6}
static patch, shear-periodic x_1	3.8×10^{-9}	1.1×10^{-7}

With the correction the remap is conservative to round-off, through regridding too (Fig. 1); the total y-momentum stays at 3×10^{-16} against 5×10^{-7} without. With shear-periodic x_1 boundaries the correction still removes most of the drift; the remaining 4×10^{-9} is the boundary remap.

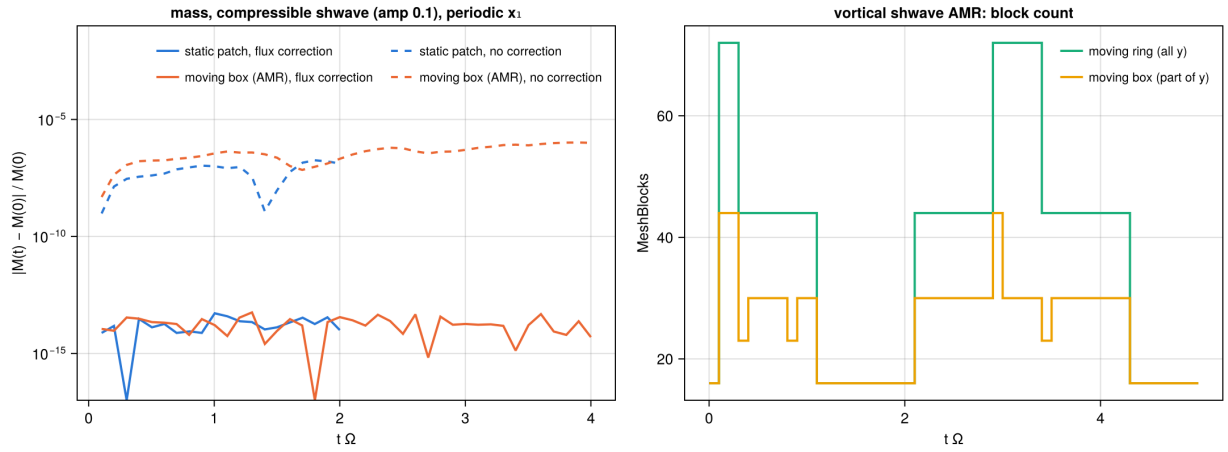


Figure 1: Left: relative mass drift of the compressible shwave (amplitude 0.1, periodic x_1) on a static patch (blue) and under a moving box with AMR (orange), with (solid) and without (dashed) the orbital-advection flux correction. Right: MeshBlock count of the vortical-shwave AMR runs of §6 — the moving box refines part of the ring's blocks. Script `figures/plot_fargo_amr_tests.jl`.

5 MPI

The conservation runs (static patch and moving box) were repeated on 3, 4 and 7 ranks. All binary dumps are bit-identical to the serial runs. The history files differ only in the volume sums, by the MPI reduction order: $\leq 6.5 \times 10^{-13}$ on a mass of 16, and $\sim 10^{-15}$ in momenta whose value is itself round-off. Mass is conserved to 1×10^{-14} on every rank count. The vortical-shwave moving box on 4 ranks is likewise bit-identical to the serial run in all 51 dumps.

6 Vortical shearing wave through static and moving level jumps

The Johnson & Gammie (2005) vortical shwave (`ipert=2`, isothermal, PPM; $64^2 \times 4$ root cells on 0.5×0.5, $16^2 \times 4$ blocks, $t = 0$–5) on five meshes, compared with a uniform 128^2 run as the reference:

run	refined region (level 1)	peak x-KE vs 128^2	max $\lvert\Delta \mathrm{KE}_x\rvert$/peak
uniform 64^2	—	-4.5%	1.1×10^{-1}
ring, static	$\lvert x\rvert < 0.125$, all y	-7.9%	8.2×10^{-2}
patch, static	$\lvert x\rvert, \lvert y\rvert < 0.125$	-8.1%	9.3×10^{-2}
moving ring, AMR	$x_c = 0.2 \sin t$	-12.2%	1.3×10^{-1}
moving box, AMR	$+\, y_c = 0.15 \sin t + 0.1t$	-11.9%	1.3×10^{-1}

Inputs: `hydro_incompress_shwave{,_smr_ring,_smr_ypatch,_amr_ring,_amr_box}`; all history output at full precision (`output1/data_format=%.15e`). The patch lies between the ring and the coarse grid in the history error, and the moving box matches the moving ring (the box creates and destroys 77 blocks, the ring 112; Fig. 1, right). The peak error of the ring is larger than that of the coarse grid; this is a property of the ring runs, not of this work.

Figure 2 maps the v_x error against the 128^2 run. Inside the static patch the error is larger than inside the ring at $t = 1$: the wave reaches the fine region after crossing coarse cells in y, while in the ring it never leaves the fine grid. At the x_2 level jumps the error bands run straight across, with no stripes or kinks. Averaged over the refined x_1 columns of the patch, the error next to the jumps lies between those of the fine interior and the coarse rows ($t = 1$: 9.1, 4.9 and 11×10^{-2} of max $\lvert v_x\rvert$; at $t = 2.5$ the three agree to 10%).

The planned twin without orbital advection is not available for patches: with `orbital_advection = false`, the patch mesh collapses at the first step ($\Delta t \to 10^{-45}$) *in the parent code as well*, with or without `prolong_primitives`. The ring mesh runs. This is a separate, pre-existing problem (§12).

7 Swing amplification with multigrid gravity on a patch

`tests/swing_selfgrav_smr_ypatch.athinput` is the swing test of `multigrid_sb_validation.pdf` (Phase 2a: $2\pi \times 2\pi \times \pi/4$, $128^2 \times 16$ root cells, 16^3 blocks, $4\pi G\rho_0 = 1.9$, a leading wave with $k_x/k_y = -6$) with the level-1 region cut to the middle four of eight block rows in y (six of eight columns in x, as for the ring). The x_2 faces at $y = \pm\pi/2$ (Fig. 4) are then level jumps for orbital advection and for the multigrid octets. The ring and uniform runs were repeated with the new code.

run	peak $d_{\cos}/A$	peak time	max $\lvert d_{\sin}\rvert/A$	peak vs uniform 256^2
uniform 128^2, multigrid	-3.511418	7.804	7.3×10^{-14}	-2.97%
ring	-3.583481	7.804	1.3×10^{-13}	-0.98%
patch	-3.532422	7.804	6.1×10^{-13}	-2.39%
uniform 256^2	-3.618801	7.804	7.7×10^{-14}	—

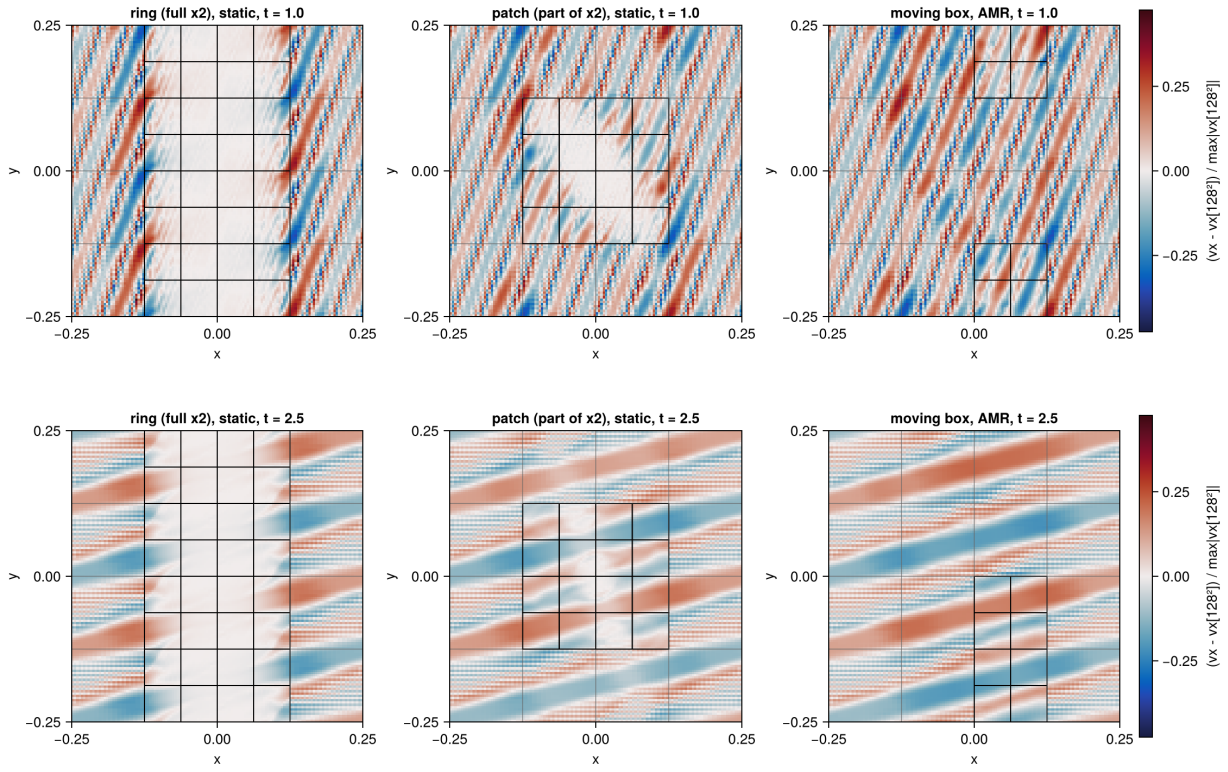


Figure 2: v_x error of the vortical shwave against a uniform 128^2 run, normalized to $\max|v_x|$, averaged over z, shown at the finest resolution (coarse cells repeated), at $t = 1$ (top) and $t = 2.5$ (bottom). Left: ring. Middle: static patch, x_2 level jumps at $y = \pm 0.125$. Right: moving box (AMR). Black: refined blocks; grey: root blocks. The checkerboard in coarse regions is the coarse cells against the fine reference. Script `figures/plot_fargo_amr_shwave.jl`.

The uniform 128^2 and ring runs reproduce the Phase-2a values to the last digit. The patch moves the peak 0.60% toward the converged answer (linear theory 3.605), under a third of the ring's 2.05%, with half the ring's refined area; the phase lock stays at machine level. The uniform 256^2 run ($256^2 \times 32$, the refined runs' finest resolution) peaks 0.38% above linear theory, at the same time and with the same machine-level phase lock. Both refined runs lie between the two uniform runs: the ring closes two thirds of the 128^2–256^2 gap in the peak (refining 75% of the area), the patch a fifth (37.5%). The gravity-off control (`hydro_srcterms/self_gravity=false`, with its own uniform 128^2 and 256^2 twins) attributes the shift as in Phase 2a: the patch-vs-uniform-128^2 deviation is larger without gravity (5.8×10^{-2} maximum, peak +1.11%) than with it (3.6×10^{-2}, +0.60%), and against the 256^2 runs the patch is again slightly farther off without gravity (3.5×10^{-2}, −2.66%) than with it (3.3×10^{-2}, −2.39%), so multigrid gravity across the x_2 level jumps adds no deviation of its own (Fig. 3).

8 The timestep limit

A cold variant of the static-patch conservation test (sound speed 0.05, amplitude 10^{-3}) lets the hydro timestep grow until the shift at $|x| = 2$ would be 8 cells against 6 available rows. The

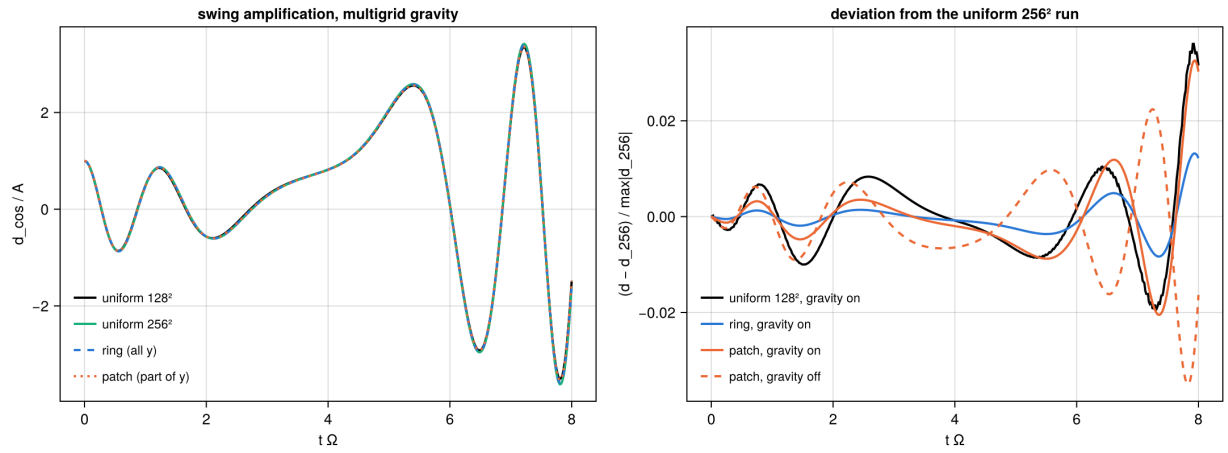


Figure 3: Swing amplification with multigrid self-gravity. Left: projection of the density onto the shearing wave, $d_{\cos}/A$, for the uniform 128^2 and 256^2, ring and patch meshes (indistinguishable at this scale). Right: deviation from the uniform 256^2 run, linearly interpolated to each run's output times, with gravity (solid) and, for the patch, without (dashed, against the gravity-off 256^2 run). Script `figures/plot_fargo_amr_tests.jl`.

build without the limit stops at the first step with the guard's message. With it, Δt is held at $0.999\,(6-2)\,\Delta x_2/(q\Omega \max|x|) = 0.0416$, exactly the predicted value; the run reaches $t = 8$ (193 steps) and conserves mass to 8×10^{-14}.

9 MHD with orbital advection on refined meshes

Branch `fargo-amr-mhd`, in two steps. First, MHD with orbital advection on refined meshes whose refinement comes in complete x_2 rings (`oa_annular = true`; the subsections up to the magnetized swing); then across x_2 level jumps (§9.5). The effective EMFs of the face-field remap are corrected at level jumps like the stage EMFs (implementation document, section "Magnetic fields: MHD on rings"). Before this, MHD + orbital advection + refinement was a fatal error, and the refined MHD shearing boxes of `multigrid_sb_validation.pdf` (Phase 4) ran without orbital advection, where the field perturbation of the magnetized swing is 13–22% off.

9.1 Regression

Uniform MHD with orbital advection (`mhd_compress_shwave`) is bit-identical to the parent code in the history and all VTK dumps: without refinement the remap's EMFs are not corrected and the arithmetic is unchanged.

9.2 The refined-mesh field error disappears with orbital advection

The three-minute reproducer of the Phase-4 field error (`inputs/tests/swing_mhd_repro.athinput`: $64^2 \times 4$ root cells, $8^2 \times 4$ blocks, $B_0 = 1$ along y, $\beta = 1.2$, static level-1 ring $|x| < \pi/4$), run with and without orbital advection. It is a magnetized shearing wave without self-gravity: the Phase-4 work found the error with and without gravity and dropped it to isolate the defect. The swing amplification with self-gravity follows in §9.4 (Fig. 5). Deviation over $t \le 6$, on common times:

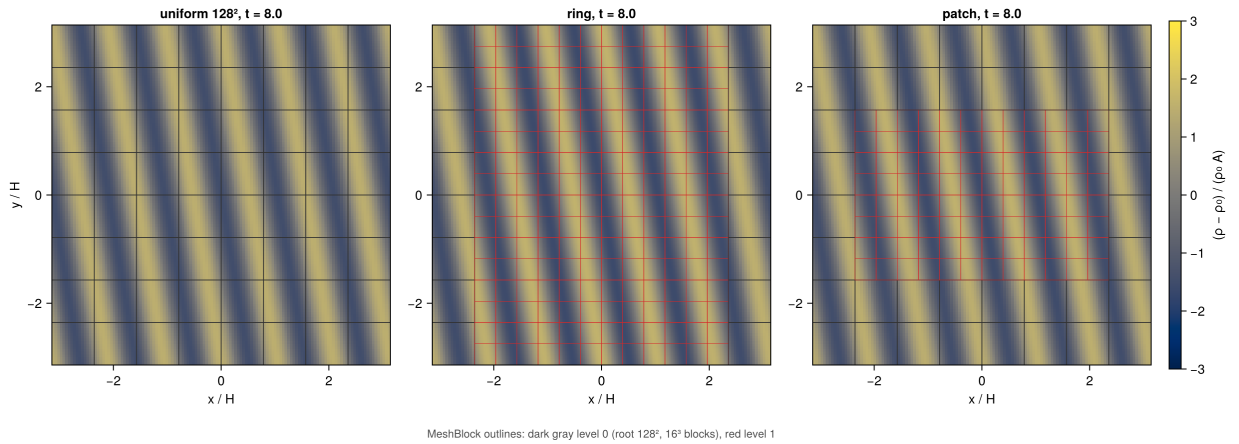


Figure 4: Where the runs of Fig. 3 are refined: midplane density perturbation $(\rho - \rho_0)/(\rho_0 A)$ at $t = 8$, near the swing peak, with the MeshBlock outlines (dark gray level 0, root 128^2 in 16^3 blocks; red level 1). The refinement is static, set by `<refined_region>` blocks, not by a criterion: the ring covers six of the eight block columns in all of y; the patch the same columns in the middle four of eight block rows, with x_2 level jumps at $y = \pm\pi/2$. Script `figures/plot_fargo_amr_meshes.jl`.

	vs linear theory		ring vs uniform twin
run	δ	c_y	c_y
uniform, no orbital advection	0.068	0.075	—
ring, no orbital advection	0.075	0.226	0.220
uniform, orbital advection	0.091	0.106	—
ring, orbital advection	0.062	0.074	0.033

With orbital advection the ring is closer to linear theory than its uniform twin, as refinement should be, and within 3.3% of it (δ: 3.0%), where the same ring without orbital advection is 22% off. The two uniform runs agree with each other to 3%. So the Phase-4 defect needs the background shear flow carried in the state across the level interfaces; with orbital advection the state holds only the perturbation and the defect does not arise. (Comparing the twins row by row of the history, as `repro.py` did, is wrong when one run's step exceeds the history interval — the uniform run with orbital advection takes $\Delta t = 0.021$ against an output interval of 0.02 — and gave a spurious 52%; `repro.py` now interpolates to common times.)

9.3 Shared faces stay consistent

The history columns `bmis`/`bnrm` measure how far the coarse copy of each face shared with a finer block is from the mean of the fine copies:

	copy mismatch / $\lvert B_n\rvert$		
mesh	EMF correction off	on	$\max \lvert\nabla\cdot B\rvert/\lvert B\rvert$
ring (level jumps at x_1 faces)	$\sim 10^{-2}$ (up to 0.7)	$\sim 10^{-15}$	4×10^{-15}
ring + half-height in z (x_1 and x_3 jumps)	median 1.8×10^{-2}	$\sim 10^{-15}$	3×10^{-15}
moving ring, AMR (280 blocks created, 224 destroyed)	median 3×10^{-3}	$\sim 10^{-15}$	5×10^{-15}
ring without orbital advection (stage EMFs only)	$\sim 10^{-15}$		4×10^{-15}

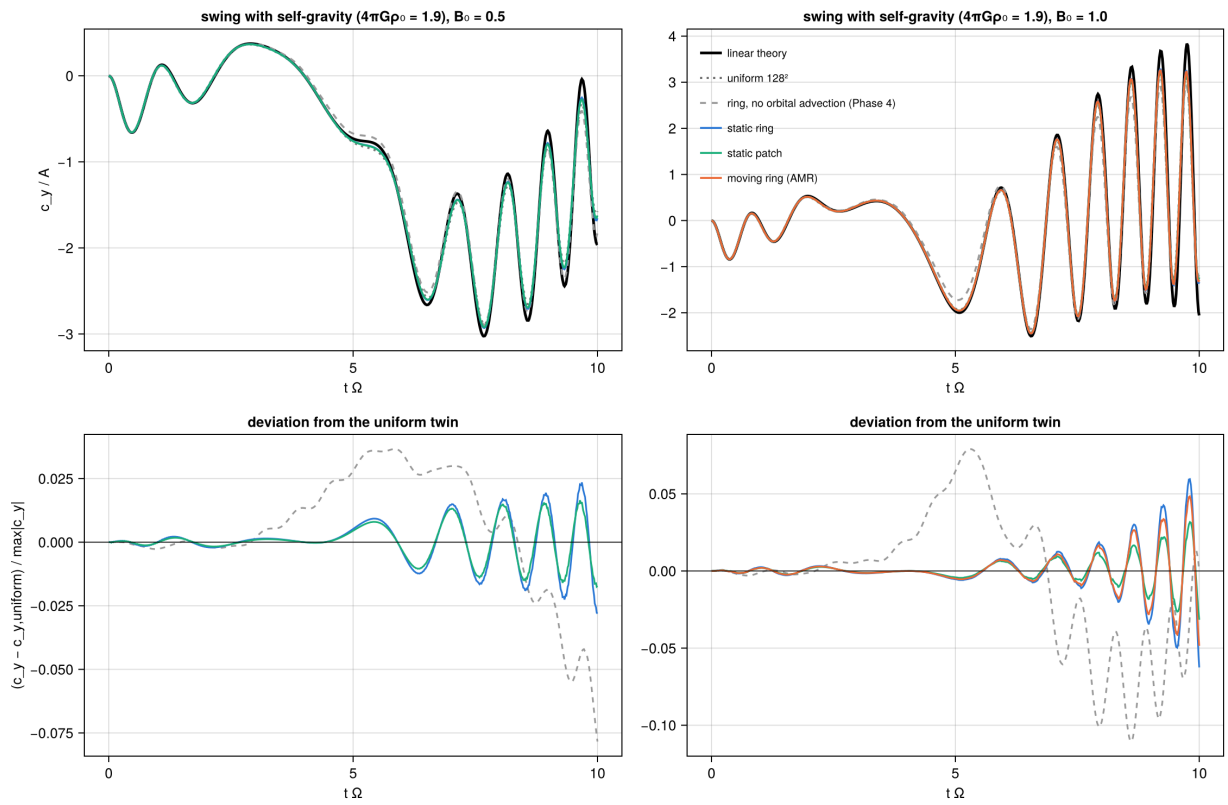


Figure 5: Magnetized swing amplification with self-gravity (`inputs/tests/swing_mhd.athinput`: $128^2 \times 16$ root, multigrid gravity with $4\pi G\rho_0 = 1.9$, amplitude 10^{-4}), $B_0 = 0.5$ (left) and 1.0 (right). Top: azimuthal field perturbation c_y against linear theory (black; `swing_ode.py`, 7-point Laplacian at the root spacing) for the uniform run and, with orbital advection, a static ring, a static patch ($|y| < \pi/2$, §9.5) and a moving ring (AMR); dashed, the Phase-4 ring without orbital advection. Bottom: deviation from the uniform twin in the same frame (orbital advection or not), normalized by the twin's maximum $|c_y|$ over $t \le 10$ (the tables use $t \le 6$). The growing drift of all runs from theory after $t \approx 6$ is numerical damping of the wave as the shear winds it to 14 cells per wavelength by $t = 10$; it converges at second order (see text). Script `figures/plot_fargo_amr_mhd.jl`.

Without the correction the two copies drift apart by about 1% of the field (the ratio peaks where B_n passes through zero); each block stays divergence-free, which is why the history's $\nabla\cdot B$ does not see it. With it they agree to round-off, as the stage's own EMF correction keeps them without orbital advection (Fig. 7). The effect on the smooth swing wave is small (2–4×10^{-4} in δ and c_y), but a shared face with two different values is a monopole of the composite mesh. On 4 MPI ranks every field dump of the x_3-jump and AMR runs is bit-identical to the serial run.

9.4 Magnetized swing amplification with gravity

The full Phase-4 test (`inputs/tests/swing_mhd.athinput`: $128^2 \times 16$ root cells, 16^3 blocks, multigrid gravity with $4\pi G\rho_0 = 1.9$, HLLD, PLM) on the two-column ring and the moving ring of Phase 4, now with orbital advection, against the uniform runs with orbital advection (`uF_b*`). Deviation from the uniform twin over $t \le 6$, with the Phase-4 values without orbital advection for comparison:

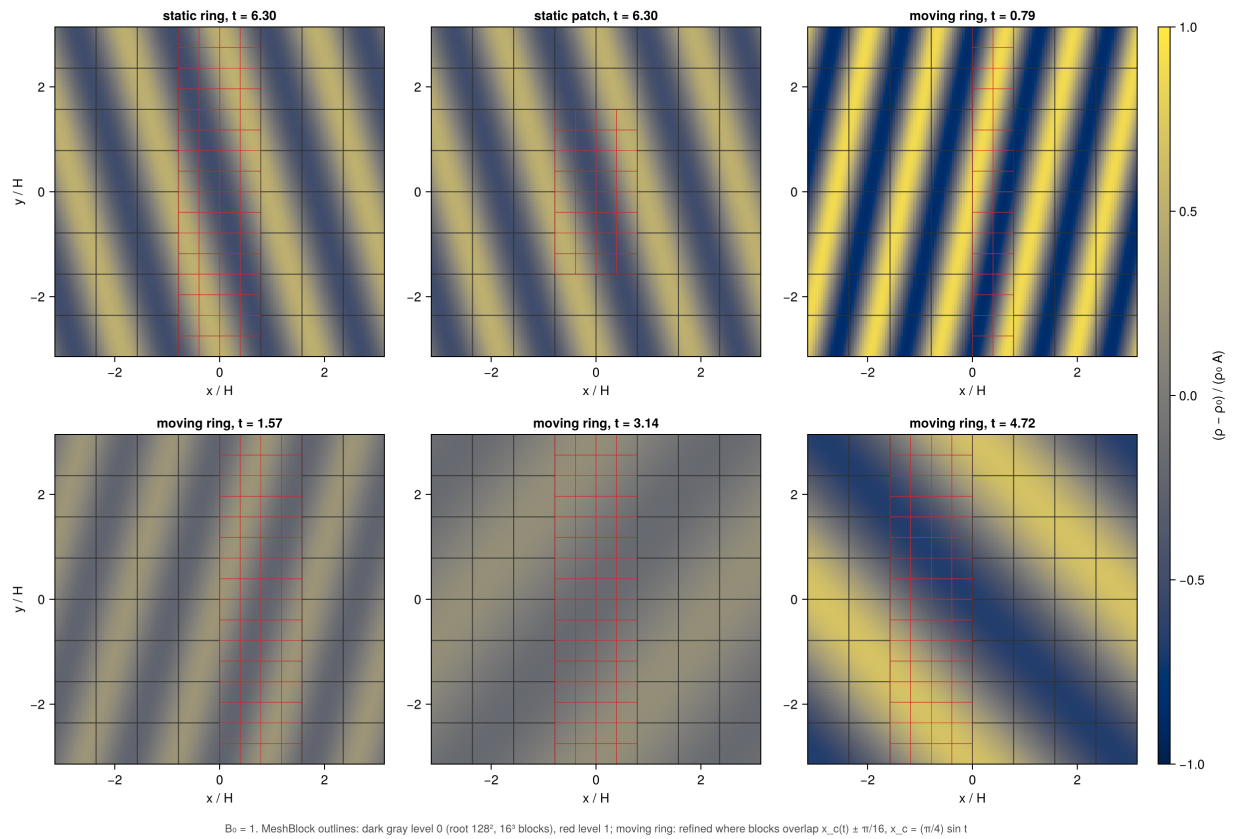


Figure 6: Where the runs of Fig. 5 are refined ($B_0 = 1$; as Fig. 4). Top: the static ring (the central two of eight block columns, $|x| < \pi/4$) and the static patch (the same columns, $|y| < \pi/2$) at $t = 6.3$, both set by `<refined_region>` blocks; the moving ring at $t = 0.79$. Bottom: the moving ring at $t = 1.57$, 3.14, 4.72. It is adaptive with a prescribed criterion (`SwingMovingRingRefine`): a block is refined when it overlaps $x_c(t) \pm \pi/16$, $x_c = (\pi/4)\sin t$, and derefined otherwise, so complete x_2 rings sweep across the centre and regrids happen at known times, independent of the flow. (These snapshots come from reruns of the same inputs with density dumps; the runs of Fig. 5 kept only histories.)

	orbital advection (this work)		without (Phase 4)	
case	δ	c_y	δ	c_y
static ring, $B_0 = 0.5$	0.035	0.017	0.025	0.071
static ring, $B_0 = 1.0$	0.014	0.014	0.039	0.133
moving ring, $B_0 = 1.0$	0.013	0.013	0.040	0.150

Figure 5 shows c_y against linear theory and the deviation from the uniform twin, together with the patch runs of §9.5; Fig. 6 shows where each mesh is refined. With orbital advection the refined runs follow their uniform twins to 1–4% in δ and 1–2% in c_y, against 7–15% in c_y without it. Against linear theory they are closer than the uniform runs, as refinement should be: c_y is off by 3.2% and 2.7% on the static rings ($B_0 = 0.5$, 1.0) and 2.9% on the moving ring, against 4.9% and 4.0% for the uniform runs. The shared faces stay consistent to round-off (copy mismatch $\leq 10^{-12}$ of $|B_n|$) and $\max|\nabla\cdot B|/|B| \leq 7 \times 10^{-15}$. The static rings (176 MeshBlocks) ran to $t = 10$; to $t \leq 8$ their c_y deviation is 1.7% and 2.6%. The moving ring (64–176 MeshBlocks; 448 created, 392 destroyed) also

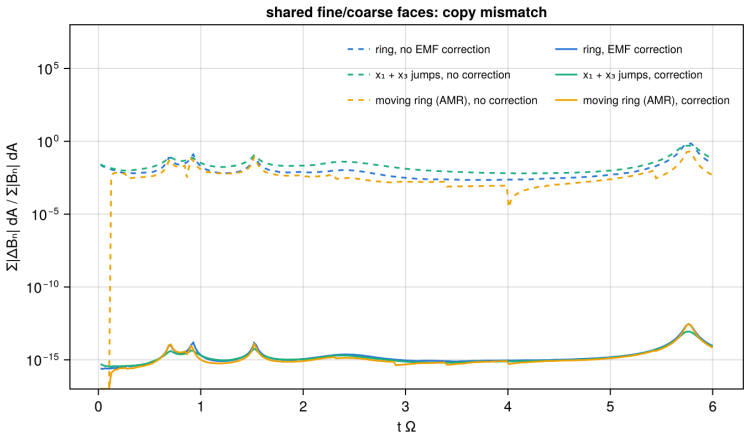


Figure 7: Mismatch of the two copies of the faces shared by fine and coarse blocks, with (solid) and without (dashed) the EMF correction of the face-field remap, in the gravity-free reproducer: ring, ring with x_3 level jumps, moving ring (AMR). Script `figures/plot_fargo_amr_mhd.jl`.

ran to $t = 10$: 1.9% and 2.3% to $t \leq 8$, with $\nabla \cdot B \leq 5 \times 10^{-15}$ throughout. All on 3 MPI ranks; inputs and runs in `validation/run/fargo_amr/mhd/swing/`.

The late drift from linear theory. After $t \approx 6$ every run in Fig. 5, uniform and refined, falls increasingly below theory. This is numerical damping of the wound-up wave, not an error of the coupling. The shear keeps winding the wave, $k_x(t) = -6 + 1.5\,t$, so at $t = 10$ its wavelength is 14 cells of the 128^2 grid, and the scheme's dissipation grows as the wave is less well resolved. The peaks of c_y after the swing, uniform runs at $B_0 = 0.5$ (Phase 4, `uF_b0.5` and `uF_b0.5_256`):

	128^2		256^2	
peak at t	cells per wavelength	$\lvert c_y\rvert/\lvert c_y^{\rm theory}\rvert$	cells per wavelength	$\lvert c_y\rvert/\lvert c_y^{\rm theory}\rvert$
6.5	33	0.969	66	0.990
7.7	23	0.955	46	0.986
8.6	18	0.933	37	0.981
9.3	16	0.893	32	0.973
10.0	14	0.822	28	0.959

The loss is in amplitude only: every peak falls within one output interval (0.02) of theory's. It converges at second order (an 18% loss at $t = 10$ becomes 4.1% at 256^2, a factor 4.4), and it is set by the resolution of the wave, not by the elapsed time: at about 32 cells per wavelength the loss is 3.1% at 128^2 ($t = 6.5$) and 2.7% at 256^2 ($t = 9.3$). $B_0 = 1.0$ behaves the same way (19–24% by $t = 9.5$–10 at 128^2). The refined runs keep most of the wave on the root grid and so damp like their uniform twins, which is why the comparison with the twin (Fig. 5, bottom) stays at the per-cent level while both drift from theory, and why the tables stop at $t \leq 6$ or 8. A shearing wave winds without end, so any fixed grid eventually under-resolves it; following one to a given $|k_x|$ with less than about 3% loss needs about 30 cells per wavelength there.

So with orbital advection, refined MHD shearing boxes are validated on rings, with and without self-gravity; the refined-mesh field error of Phase 4 remains a property of the full-shear-flow frame.

9.5 Across x_2 level jumps

In the second step the face-field remap takes x_2 neighbours one level coarser or finer (implementation document), and MHD with orbital advection follows the hydro policy: `oa_annular` is optional.

Regression. Uniform MHD with orbital advection is still bit-identical to the parent code, and the ring, x_3-jump and moving-ring reproducer runs of the first step are bit-identical to their first-step results: the same-level path is unchanged.

Reproducer on a patch. The reproducer with the ring cut to four of eight block rows in y ($|x| < \pi/4$, $|y| < \pi/2$; 120 MeshBlocks), and with a moving box under AMR (the moving ring of the first step with an x_2 window $y_c = \sin t + 0.5\,t$, half-width 0.6; 64–106 MeshBlocks, 147 created, 126 destroyed):

mesh	vs uniform twin δ	vs uniform twin c_y	copy mismatch / $\lvert B_n\rvert$ correction on / off	$\max \lvert\nabla\cdot B\rvert/\lvert B\rvert$
ring (first step)	0.030	0.033	10^{-15} / 10^{-2}	4×10^{-15}
static patch (x_2 jumps)	0.025	0.026	2×10^{-15} / 6×10^{-7}	4×10^{-15}
moving box, AMR	0.023	0.023	10^{-15} / 2×10^{-7}	4×10^{-12} (see below)

On the patch the field perturbation follows its uniform twin as closely as on the ring, and every shared face, now including the x_2 faces, stays consistent to round-off. (Without the correction the normalized mismatch is smaller than on the ring only because the x_2 faces add their normal field $B_y \approx B_0 = 1$ to the normalization; the absolute mismatch is of the same order.) On 4 MPI ranks every field dump of both runs is bit-identical to the serial run.

A pre-existing $\nabla\cdot B$ defect at AMR derefinement. In the moving-box run $|\nabla\cdot B|/|B|$ grows in steps, 10^{-13} to 5×10^{-12}, at the regrid events where blocks are *derefined*, and stays constant in between (so the remap and its correction conserve it). Two controls place it outside this work: the same moving box with $q = 0$ (the remap then shifts nothing) and with $q = 0$ and orbital advection off give the same jumps at the same events (3.9×10^{-13} at $t = 0.28$, 7.1×10^{-12} by $t = 3$). The moving ring of the first step derefines too (392 blocks destroyed in the full swing run) and stays at 5×10^{-15}; a box adds fine neighbours across x_2 faces and x_1x_2 edges to a derefined block. The suspect is AthenaK's post-AMR face-field handling (`MeshRefinement::RepairAMRFC`, which recomputes the interior faces of created *and* derefined blocks, or the face ownership of the receives at those neighbours). It is small now but accumulates with every derefinement; it is left for a separate fix.

Magnetized swing with gravity on a patch. The full test of §9 (multigrid gravity, $128^2 \times 16$, 3 MPI ranks) with the two-column ring cut to half the azimuth ($|y| < \pi/2$):

case	vs uniform twin, $t \le 6$ δ	vs uniform twin, $t \le 6$ c_y	vs theory, $t \le 6$ c_y	$\max \lvert\nabla\cdot B\rvert/\lvert B\rvert$
patch, $B_0 = 0.5$	0.030	0.015	0.035	5×10^{-15}
patch, $B_0 = 1.0$	0.012	0.011	0.030	6×10^{-15}
ring, $B_0 = 0.5$ (above)	0.035	0.017	0.032	6×10^{-15}
ring, $B_0 = 1.0$ (above)	0.014	0.014	0.027	7×10^{-15}

The patch (120 MeshBlocks) follows its uniform twin as closely as the ring, 1–3% in δ and c_y, also to $t \leq 8$ (1.4%, 1.7% at $B_0 = 1.0$), and linear theory better than the uniform runs (c_y: 3.5% and 3.0% against 4.9% and 4.0%). Shared faces agree to 5×10^{-15} of $|B_n|$, and $\nabla \cdot B$ stays at round-off. Runs in `validation/run/fargo_amr/mhd/swing/` (`ypF_b*`, input `ypatch.athinput`) and `mhd/stepB/` (reproducer).

10 GPU

The same inputs on one NVIDIA H200 (h-gpu; CUDA 13.3, Kokkos Hopper build, no MPI), October 28, 2026. The CUDA compiler found one defect the CPU build accepts: an extended device lambda in a private member function (`OrbitalAdvectionFC::UpdateB`), now public. Then all thirteen runs (seven hydro, six MHD; 31 s in total) ran to the same cycle counts as on the CPU:

runs	GPU vs CPU (max over t)	checks on the GPU
hydro: conservation, Δt limit, epicycle, shwave patch and box	$\leq 8 \times 10^{-13}$	mass $\leq 10^{-14}$ with correction, 1.8×10^{-7} without
MHD: ring, ring without correction, patch, uniform	$\leq 2 \times 10^{-12}$	face mismatch $\leq 3 \times 10^{-13}$, $\nabla \cdot B \leq 4 \times 10^{-15}$
MHD: x_3 jumps, moving box	$\leq 7 \times 10^{-9}$	face mismatch $\leq 10^{-13}$; box $\nabla \cdot B$ 2×10^{-12} (§9.5)

(GPU vs CPU: physical history columns — mass, energies, wave projections — relative to their maxima.) Repeated GPU runs of the two last cases are bit-identical, so the larger GPU–CPU difference there is round-off growing differently (limiter branches at near-zero differences), not a race. Runs in `h-gpu:/data/limjay/runs/fargo_amr_gpu/`, scripts `build_cuda.slurm`, `run_tests.slurm`.

11 Nonlinear swing to collapse

The swing tests so far are linear. This section pushes the swing of `multigrid_sb_validation.pdf` (§13, Figs. 9–11 there) into collapse, with the refinement now free to follow the clump as a patch. Setup (`inputs/tests/swing_selfgrav_clump_jeans.athinput`): $2\pi \times 2\pi \times \pi/4$, $128^2 \times 16$ root cells, 16^3 blocks, isothermal, HLLE, PLM, multigrid gravity with $4\pi G\rho_0 = 6$, a leading wave $k_x/k_y = -6$ of amplitude 0.2, orbital advection on, to $t = 10$. At these parameters the uniform run of §13 fragments at $t \approx 4.5$ and forms a bound clump near $t \approx 9$. In §13 the zoom was one level, triggered by density ($\rho > 2$), and had to come in complete x_2 rings. Here it is the Jeans criterion of the implementation document (`SwingJeansRefine`): a block is refined where $\lambda_J = 2\pi c_s/(4\pi G\rho)^{1/2}$ is shorter than 16 of its cells, up to three levels, and derefined where λ_J exceeds 40 cells everywhere in it. λ_J is 52 root cells at ρ_0, so the levels switch on at $\rho = 10.7$, 43 and 171, and level 3 keeps $\lambda_J \geq 16$ cells up to $\rho = 683$. The 128^2 Truelove limit ($\lambda_J = 4$ cells) is 171. All runs on one H200 (h-gpu), October 28, 2026.

Hydro. Uniform 128^2, the same with x_2 rings (`oa_annular = true`), with patches (`oa_annular = false`), uniform 256^2, and a marginal case ($4\pi G = 5$, amplitude 0.05, patches) that should not collapse:

run	peak ρ/ρ_0 (at t)	first $\rho > 100$	max blocks	MeshBlock-cycles	wall time
uniform 128^2	203 (9.40)	9.30	64	1.2×10^5	1.6 min
uniform 256^2	40.4 (10.0)	—	512	1.5×10^6	5.4 min
AMR, rings	286 (9.50)	9.20	4768	1.3×10^7	57 min
AMR, patches	377 (10.0)	9.15	610	1.5×10^6	23 min
marginal, patches	2.70 (5.65)	—	64	5.6×10^4	0.8 min

The uniform 128^2 run reproduces §13 on the GPU: peak $203.28\rho_0$ at $t = 9.40$, the CPU value to the printed digits.

Patches and rings give the same collapse. Both refined runs step through the same first crest (peak $25.62\rho_0$ at $t = 4.65$, level 1 only), cross $\rho = 10.7$ again at $t = 8.20/8.25$ and $\rho = 100$ at 9.15/9.20, and collapse at the same site (Fig. 9). Their winding-wave amplitudes $d_{\cos}/A$ differ by at most 0.012 up to $t = 7$ and 0.08 up to $t = 8$, against 0.13 between either and the uniform run already at $t \leq 7$ (Fig. 8, middle): the choice of patch or ring is ten times smaller than the effect of refinement itself, until the collapse turns chaotic ($t \gtrsim 8.5$; afterwards peak densities differ by up to a factor 2 between any two runs, as the clump oscillates).

Refinement moves the clump to where the resolved runs of §13 put it. The uniform 128^2 run ends with one clump at the box centre ($131\rho_0$ at $(0.02, -0.02)$, peak 203 at $t = 9.4$, above its Truelove limit). Both refined runs collapse instead at $y = \pm\pi$, as a pair of knots related by the inversion symmetry of the initial wave (patches at $t = 9.5$: $161.6\rho_0$ at $(-0.26, -2.65)$ and $(0.26, 2.65)$), one of which wins by $t = 10$ ($369\rho_0$ on patches, $206\rho_0$ on rings, in the midplane). That is the site of the dominant knot of the uniform 256^2 multigrid run and of its FFT twin in §13, which identified the 128^2 central clump as the under-resolved outlier. The uniform 256^2 run has not collapsed by $t = 10$: it holds a thin filament along y at $x \approx 0$, densest near $y = \pm\pi$ ($40.4\rho_0$ at $(\mp 0.01, \mp 2.88)$, the §13 value), where its extension to $t = 14$ in §13 forms the dominant knot.

But refinement does not fix the clock. The 256^2 run collapses about one time unit later: at $t = 8.5$ its peak is $1.9\rho_0$, against 20–$22\rho_0$ in the refined runs and $8.4\rho_0$ in the uniform 128^2 run. Between the crests ($t = 5.2$ to 8.2) every block of the refined runs is back at level 0 — the gas is at $\rho \lesssim 4$, where λ_J is 26 root cells or more — so for those three time units they are 128^2 runs. The winding wave says the same: against the 256^2 run, $d_{\cos}/A$ of the refined runs is off by 0.14–0.19 up to $t = 7$ (the uniform 128^2 run by 0.27–0.31), and all three part from it at $t \approx 8$. The Jeans criterion at 16 cells resolves gravity; it does not resolve the hydrodynamics of the wound-up crests at $\rho \approx 2$–4, which still differ between 128^2 and 256^2. A converged collapse time needs a finer root grid or a criterion that also refines the crests; neither was run.

The clump stays resolved. The peak stays below 683, so level 3 keeps $\lambda_J \geq 16$ cells to the end; the uniform 128^2 run overshoots its Truelove limit (203 against 171).

Cost. Rings refine the clump's x_1 columns at all y: at $t = 9.5$, 4096 level-3 blocks (eight rings of 512) to hold a clump a few level-3 blocks across. Patches need 610 at most, a factor 8.6 fewer MeshBlock-cycles over the run. On the GPU the wall time drops only by 2.5: a 600-block mesh does not fill an H200, and the AMR machinery (regridding every five cycles, multigrid across levels, the orbital-advection messages per sub-face) costs per block. Per MeshBlock-cycle the patch run takes 0.90 ms, the ring run 0.26 ms, the uniform 256^2 run 0.21 ms — which therefore finishes in a quarter of the patch run's wall time with the same number of MeshBlock-cycles. On CPUs, where the cost follows the block count, patches gain the full factor over rings.

The marginal case. At $4\pi G = 5$, amplitude 0.05, the crest peaks at $2.70\rho_0$ ($t = 5.65$) and re-expands, as in §13; λ_J never falls below 16 cells, no block is refined, and the run is the uniform run (the same peak to seven digits as the CPU run of §13).

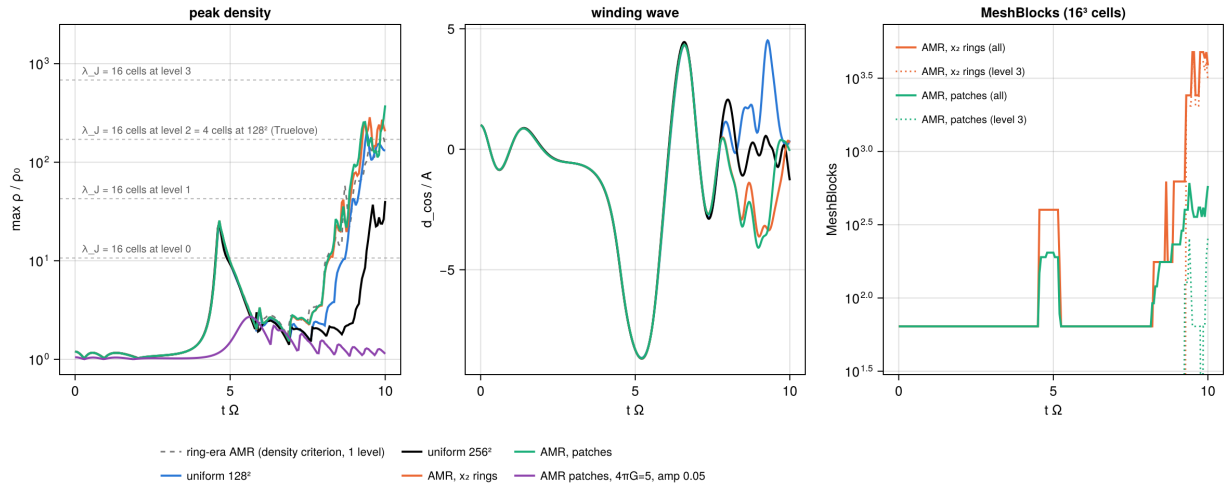


Figure 8: Nonlinear swing to collapse, hydro ($4\pi G = 6$, amplitude 0.2). Left: peak density, with the densities at which λ_J falls to 16 cells of levels 0–3 (the refinement thresholds), the ring-era run of §13 (one level, density criterion) and the marginal case. Middle: winding-wave amplitude; the refined runs part from the uniform 128^2 run after the first crest ($t \approx 5$) and from each other at $t \approx 8.5$. Right: MeshBlock count, total (solid) and level 3 (dotted). Script `figures/plot_fargo_amr_clump.jl`.

MHD. The same with a uniform field along y (`inputs/tests/swing_mhd_clump_jeans.athinput`; isothermal HLLE — isothermal HLLD is unstable at $\beta \lesssim 1$ — and the thermal Jeans length, so the criterion ignores magnetic support): $B_0 = 0.5$ ($\beta = 8$) on the three 128^2 meshes and uniform 256^2 (to $t = 12$), and $B_0 = 0.25$ uniform and on patches:

run	first crest	$\rho > 100$	$\rho > 683$	max ρ/ρ_0	MeshBlock-cycles	wall time
uniform, $B_0 = 0.5$	38.0 (4.45)	—	—	38	9.5×10^4	1.4 min
rings, $B_0 = 0.5$	36.8 (4.45)	9.70	9.90	1.4×10^4	7.3×10^6	34 min
patches, $B_0 = 0.5$	36.8 (4.45)	9.60	9.75	1.1×10^7	1.6×10^6	23 min
uniform 256^2, $B_0 = 0.5$	36.3 (4.45)	10.90	11.25	1.4×10^5	2.5×10^6	9.6 min
uniform, $B_0 = 0.25$	2.46 (5.65)	—	—	20.4	9.1×10^4	1.3 min
patches, $B_0 = 0.25$	2.46 (5.65)	—	—	20.4	8.6×10^4	1.4 min

At $B_0 = 0.5$ rings and patches again agree through the swing and the first crest (36.769 and $36.773\rho_0$ at $t = 4.45$, level 1), with winding waves within 0.044 of each other up to $t = 7$ and 0.13 up to $t = 8$, against 0.25 between either and the uniform run already at $t \leq 7$. They cross $\rho = 10.7$ again at $t = 8.70/8.75$ and run away near the box centre, the patch run 0.1 earlier (Figs. 10 and 11).

The uniform runs collapse later, and elsewhere. The uniform 128^2 run has not collapsed by $t = 10$ (peak $17.4\rho_0$, rising); the uniform 256^2 run collapses at $t = 10.9$ ($\rho > 100$), 1.2–1.3 time units after the refined runs, and runs away to $1.4 \times 10^5\rho_0$ by $t = 11.85$, past its own Truelove limit (683). The order is the hydro order: more resolution, later collapse, with the refined runs earliest; their winding wave departs from the 256^2 run sooner than the uniform 128^2 run's does (0.6 against 0.4 in $d_{\cos}/A$ by $t = 7$). Both uniform runs collapse on the shear-periodic x_1 boundary — the 256^2 clump as an inversion-symmetric pair at $(\pm 3.13, \pm 1.56)$, the 128^2 peak at $(3.12, -3.02)$ — while both refined runs collapse at the centre. Invariant I1 forbids refinement on the x_1 boundary, so the refined runs keep the boundary site at 128^2 and refine the central one, which then wins. In hydro all

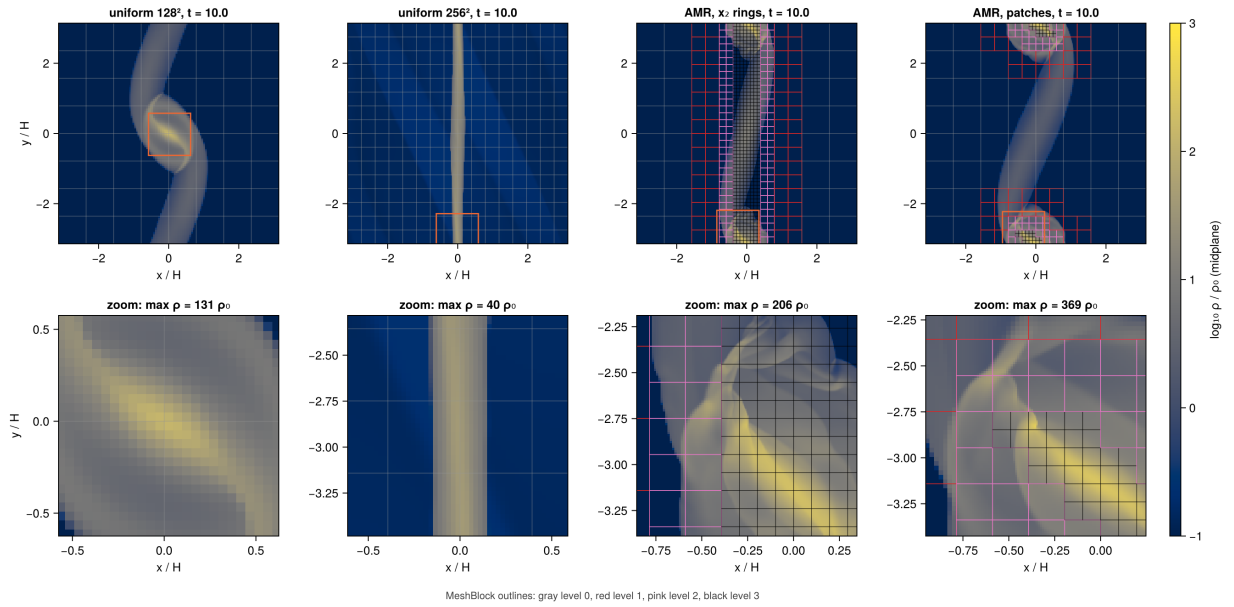


Figure 9: Midplane density at $t = 10$ with MeshBlock outlines (top) and a 1.2×1.2 zoom centred on each run's densest cell (bottom; the box is periodic in y, so the zoom may cross $y = \pm\pi$). The refined runs collapse at $y = \pm\pi$, where the 256^2 filament is densest; the uniform 128^2 run at the centre.

runs collapse near $x \approx 0$ and this does not arise; in MHD the refined runs are wrong in both the time and the place of the collapse, and the place is set by the mesh policy. Following a clump that forms on the x_1 boundary needs refinement there, i.e. lifting I1 (shear-periodic exchange between faces at different levels), beyond this work.

Unlike the hydro clump, which oscillates at 200–400ρ_0, the magnetized clump of the refined runs runs through the level-3 Jeans limit (683) and the level-3 Truelove limit (1.1×10^4) within 0.25 time units: with `num_levels = 4` and no sink, the last 0.1–0.2 of these runs is unresolved, and the two meshes differ there in shape (a symmetric pair of knots on rings, a central clump with two arms on patches, Fig. 11). At $B_0 = 0.25$ the first crest is weak (2.46); the second forms at the shear-periodic x_1 boundary ($x = -3.12$, $t = 9.2$, $20.4\rho_0$), where invariant I1 forbids refinement, so the patch run never refines. It follows the uniform run to 0.006 in $d_{\cos}/A$ up to $t = 7$ and reaches the same peak, but is not bit-identical: the two differ at the 10^{-6} level from the first steps (with refinement enabled the run takes other code paths, e.g. in the multigrid solver; not traced), and the second crest amplifies this to 0.2 by $t = 8$. Each block stays divergence-free to round-off ($|\nabla \cdot B|\Delta x/|B| \leq 8 \times 10^{-15}$, volume-weighted) in every run.

Shared faces at two and more levels. The shared-face mismatch stays at round-off ($\leq 5 \times 10^{-15}$) as long as one level is present, but jumps to 10^{-3}–7×10^{-3} when level 2 appears ($t = 9.5$ on patches, 9.65 on rings). A small reproducer isolates it (`run/fargo_amr/mhd/twolev`: the moving-box input of §9.5 with `num_levels = 3`, $t \leq 3$):

run	mismatch, max (median)	max $\lvert\nabla\cdot B\rvert/\lvert B\rvert$
static, two levels (nested patches)	5×10^{-15}	4×10^{-15}
moving box, one level	1.4×10^{-15}	1.4×10^{-12}
moving box, two levels	7×10^{-7} (2×10^{-8})	5×10^{-11}
the same, $q=0$	5×10^{-7} (3×10^{-8})	4×10^{-11}
the same, $q=0$, no orbital advection	5×10^{-7} (3×10^{-8})	4×10^{-11}
the same, $q=1.5$, no orbital advection	1.2×10^{-3} (2×10^{-5})	5×10^{-8}

The mismatch changes only at regrid events and is constant between them (the EMF corrections, of the stage and of the remap, preserve it); it disappears when a later regrid removes the faces. It is the same with no shear and no orbital advection, and largest without orbital advection, so it is AthenaK's handling of face fields at AMR regrids with two or more levels — the defect behind the $\nabla\cdot B$ jumps at derefinement of §9.5, here seen on the shared faces — not the remap. In the collapse runs it arises only after $t=9.5$, when the clumps are already running away.

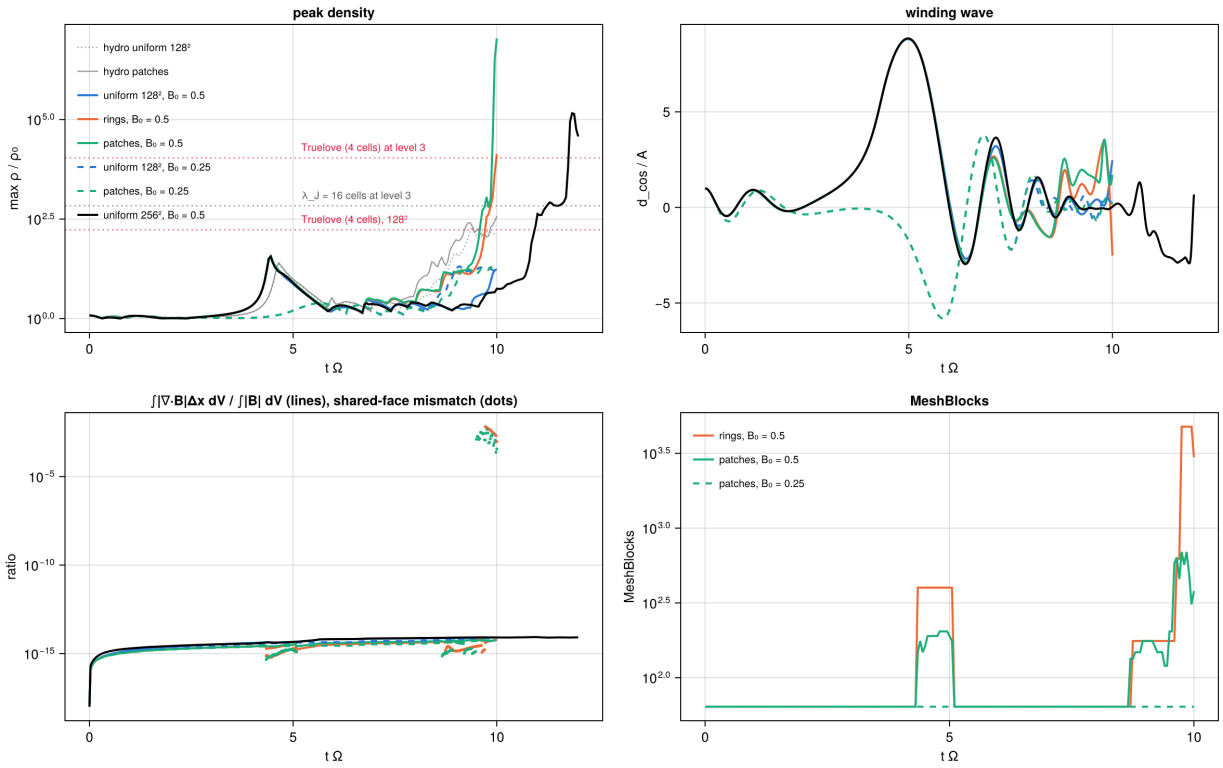


Figure 10: Nonlinear swing, MHD. Top: peak density (hydro runs in gray for reference) and winding-wave amplitude. Bottom left: $\lvert\nabla\cdot B\rvert$ (lines) and the shared-face mismatch (dots); the jump at $t\approx9.5$ comes with level 2 (see text). Bottom right: MeshBlock count.

12 Not yet validated

- **SC14 production.** The clump-following AMR of the gravito-turbulent box (`gravito_turb_l64z16_clump_amr_overlay.athinput`) restarted from an archived $\beta=3$ state, once with `oa_annular = true` and once without: block count, cost per cycle and whether the timestep limit binds. Needs a cluster.

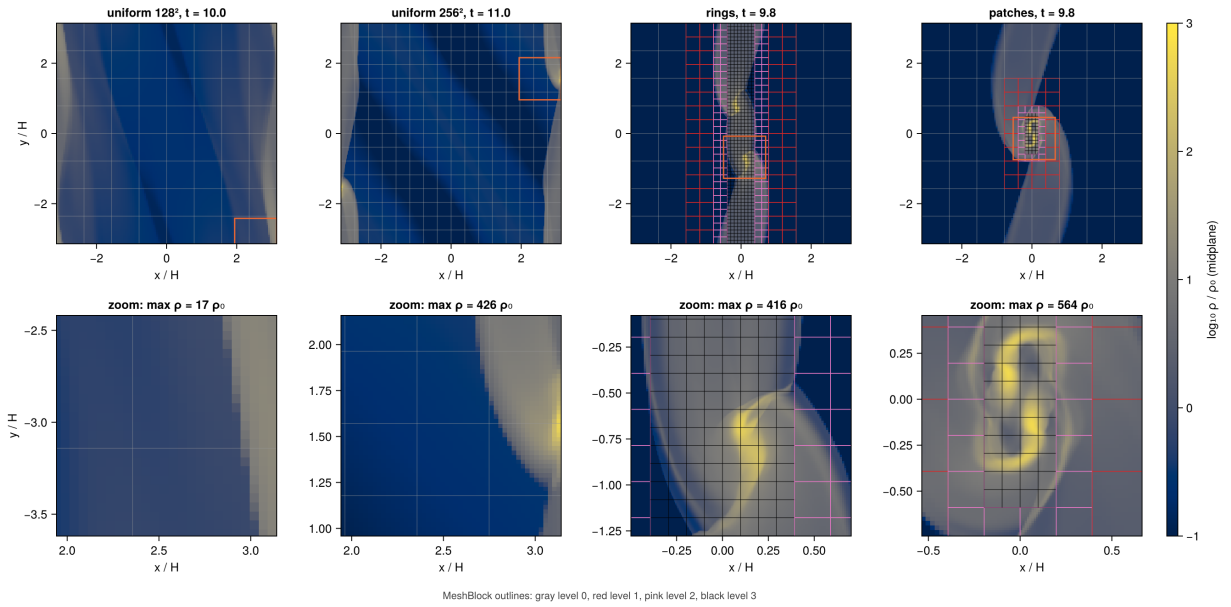


Figure 11: Midplane density of the $B_0 = 0.5$ runs, as Fig. 9: the refined runs at $t = 9.8$ and the uniform 256^2 run at $t = 11$ (each near $400\rho_0$), the uniform 128^2 run at $t = 10$. The uniform runs collapse on the shear-periodic x_1 boundary (the zooms stop at $x = \pi$), the refined runs at the centre.

- $\nabla\cdot B$ **at AMR derefinement** next to fine x_2 neighbours (§9.5): pre-existing in AthenaK, 10^{-13}–10^{-12} per event, to be fixed separately. Production MHD runs with box-shaped AMR should watch the history's `divb`.
- **Face fields at AMR regrids with two or more levels** (§11): shared faces differ by 10^{-7}–10^{-3} after regrids, with or without orbital advection; reproducer in `run/fargo_amr/mhd/twolev`. Probably the same root as the derefinement $\nabla\cdot B$ jumps.
- **Converged collapse.** Refined runs collapse 1–1.3 time units before the uniform 256^2 runs, hydro and MHD; a finer root grid or a criterion that refines the crests is needed, and a 512^2 run to check 256^2.
- **Refinement on the shear-periodic x_1 boundary** (invariant I1). The 256^2 MHD clump forms there, out of reach of the refined runs.
- **Non-FARGO refinement with x_2 level jumps** fails in the parent code (§6); not addressed here.

13 Reproduction

```
cd validation/run/fargo_amr        # README.md: one command per run directory
python3 compare.py unif128 unif ring ypatch amrring amrbox   # shwave table
python3 errmap.py 00010                                      # error at the level jumps
(cd swing && python3 cmp.py)                                 # swing table
python3 ../swing_mhd_repro/repro.py 6 base fargo fargo_nofc  # MHD reproducer pairs
cd ../.. && julia figures/plot_fargo_amr_shwave.jl           # Fig. 2
julia figures/plot_fargo_amr_tests.jl                        # Figs. 1 and 3
julia figures/plot_fargo_amr_mhd.jl                          # MHD swing figures
```

```
julia figures/plot_fargo_amr_meshes.jl       # refined regions (run/fargo_amr/meshfig)
julia figures/plot_fargo_amr_clump.jl     # nonlinear swing (run/fargo_amr/clump; README)
```

A Linear theory of the swing tests

The linear swing tests of §7 and §9 (Figs. 3 and 5) are compared with the amplitude equations of a single shearing wave in a self-gravitating, magnetized shearing sheet. This appendix derives them, states the form the code solves (the discrete Laplacian), and lists the checks. The equations are those of the problem generator's header (`src/pgen/tests/swing.cpp`) and are integrated by `validation/run/swing_mhd/swing_ode.py`.

A.1 Background

Local Cartesian coordinates rotating at Ω: x radial, y azimuthal, z vertical. The equilibrium is a uniform density ρ_0, pressure p_0, the shear flow $\mathbf{v}_0 = -q\Omega x\,\hat{\mathbf{y}}$ (with $q = 3/2$ for Keplerian rotation, epicyclic frequency $\kappa^2 = 2(2-q)\Omega^2$), and a uniform azimuthal field $\mathbf{B}_0 = B_0\hat{\mathbf{y}}$. Units are Heaviside–Lorentz, as in AthenaK: the magnetic pressure is $B^2/2$ and the Alfvén speed $v_A = B_0/\rho_0^{1/2}$. The field is in equilibrium with the shear, since $\mathbf{v}_0 \times \mathbf{B}_0 = 0$, and the tidal force balances the Coriolis force of $\mathbf{v}_0$. The uniform background has no self-gravity of its own (the Jeans swindle; in the periodic box the mean density does not source the potential).

The equations are

$$\partial_t \rho + \nabla\cdot(\rho\mathbf{v}) = 0, \tag{1}$$

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{\nabla p}{\rho} - \nabla\Phi + \frac{(\nabla\times\mathbf{B})\times\mathbf{B}}{\rho} - 2\Omega\hat{\mathbf{z}}\times\mathbf{v} + 2q\Omega^2 x\,\hat{\mathbf{x}}, \tag{2}$$

$$\partial_t \mathbf{B} = \nabla\times(\mathbf{v}\times\mathbf{B}), \qquad \nabla^2\Phi = 4\pi G(\rho-\rho_0). \tag{3}$$

The perturbations are isentropic, $\delta p = c_s^2\,\delta\rho$: isothermal runs use the isothermal sound speed, ideal-gas runs $c_s^2 = \gamma p_0/\rho_0$, with the initial pressure perturbation set to match. All the linear tests here use $c_s = 1$.

A.2 Linearized equations

Write $\rho = \rho_0 + \delta\rho$, $\mathbf{v} = \mathbf{v}_0 + \mathbf{u}$, $\mathbf{B} = \mathbf{B}_0 + \mathbf{b}$, $\Phi = \delta\Phi$, and restrict to perturbations independent of z with $u_z = b_z = 0$ (they stay zero: nothing couples to them at $k_z = 0$). With the Lagrangian derivative of the shear flow $\mathcal{D} \equiv \partial_t - q\Omega x\,\partial_y$ and the terms $(\mathbf{u}\cdot\nabla)\mathbf{v}_0 = -q\Omega u_x\hat{\mathbf{y}}$ and $-2\Omega\hat{\mathbf{z}}\times\mathbf{u} = (2\Omega u_y, -2\Omega u_x)$:

$$\mathcal{D}\frac{\delta\rho}{\rho_0} = -\nabla\cdot\mathbf{u}, \tag{4}$$

$$\mathcal{D}u_x = 2\Omega u_y - c_s^2\partial_x\frac{\delta\rho}{\rho_0} - \partial_x\delta\Phi + \frac{B_0}{\rho_0}\left(\partial_y b_x - \partial_x b_y\right), \tag{5}$$

$$\mathcal{D}u_y = -(2-q)\Omega u_x - c_s^2\partial_y\frac{\delta\rho}{\rho_0} - \partial_y\delta\Phi. \tag{6}$$

The Lorentz force is linearized as $[(\mathbf{B}_0\cdot\nabla)\mathbf{b} - \nabla(\mathbf{B}_0\cdot\mathbf{b})]/\rho_0$; its y component, $B_0(\partial_y b_y - \partial_y b_y)/\rho_0$, vanishes: a field along y exerts no force along itself at $k_z = 0$. For the induction equation, with $\nabla\times(\mathbf{a}\times\mathbf{c}) = \mathbf{a}(\nabla\cdot\mathbf{c}) - \mathbf{c}(\nabla\cdot\mathbf{a}) + (\mathbf{c}\cdot\nabla)\mathbf{a} - (\mathbf{a}\cdot\nabla)\mathbf{c}$,

$$\nabla\times(\mathbf{v}_0\times\mathbf{b}) = (\mathbf{b}\cdot\nabla)\mathbf{v}_0 - (\mathbf{v}_0\cdot\nabla)\mathbf{b} = -q\Omega b_x\hat{\mathbf{y}} + q\Omega x\,\partial_y\mathbf{b}, \tag{7}$$

$$\nabla\times(\mathbf{u}\times\mathbf{B}_0) = B_0\,\partial_y\mathbf{u} - B_0(\nabla\cdot\mathbf{u})\,\hat{\mathbf{y}}, \tag{8}$$

so that

$$\mathcal{D}b_x = B_0\,\partial_y u_x, \qquad \mathcal{D}b_y = -q\Omega b_x - B_0\,\partial_x u_x. \tag{9}$$

The first term of the b_y equation is the shear stretching radial field into azimuthal field; the second, compression of the azimuthal field by radial motion.

A.3 Shearing waves

The coefficients depend on x only through $\mathcal{D}$, so the equations admit shearing waves (Kelvin modes) with a time-dependent radial wavenumber,

$$\theta = k_x(t)\,x + k_y y, \qquad k_x(t) = k_{x0} + q\Omega k_y t. \tag{10}$$

Acting on $f(t)\,e^{i\theta}$, ∂_t gives $\dot f + i\dot k_x x f$ and $-q\Omega x\partial_y$ gives $-iq\Omega k_y x f$; the two cancel, so $\mathcal{D}$ becomes the time derivative of the amplitude. A wave with $k_{x0} < 0$ is leading; the shear winds it through $k_x = 0$ at $t = -k_{x0}/(q\Omega k_y)$ (here $t = 4$ for $k_{x0} = -6$, $k_y = 1$) into a trailing wave. In the shear-periodic box the wave is an exact solution for all t when k_{x0} and k_y are box wavenumbers, because the boundary shift $q\Omega L_x t$ in y matches the change of k_x.

Take the phases of the initial condition (a density wave at rest):

$$\frac{\delta\rho}{\rho_0} = \delta(t)\cos\theta, \quad \mathbf{u} = \mathbf{w}(t)\sin\theta, \quad \mathbf{b} = \mathbf{c}(t)\cos\theta, \quad \delta\Phi = \phi(t)\cos\theta. \tag{11}$$

Then $\nabla\cdot\mathbf{u} = (k_x w_x + k_y w_y)\cos\theta$, $\partial_x(\delta\rho/\rho_0) = -k_x\delta\sin\theta$, $\partial_y b_x - \partial_x b_y = -(k_y c_x - k_x c_y)\sin\theta$ and $\partial_y u_x = k_y w_x\cos\theta$, and the phases match term by term. The Poisson equation gives $\phi = 4\pi G\rho_0\,\delta/D$ with $D = -k^2 = -(k_x^2 + k_y^2)$ in the continuum. Collecting,

$$\boxed{\begin{aligned} \dot\delta &= -(k_x w_x + k_y w_y), \\ \dot w_x &= 2\Omega w_y + k_x\Psi\,\delta - \frac{B_0}{\rho_0}\,(k_y c_x - k_x c_y), \\ \dot w_y &= -(2-q)\Omega w_x + k_y\Psi\,\delta, \\ \dot c_x &= k_y B_0\,w_x, \\ \dot c_y &= -k_x B_0\,w_x - q\Omega c_x, \end{aligned} \qquad \Psi(t) = c_s^2 + \frac{4\pi G\rho_0}{D(t)},} \tag{12}$$

with $\delta(0) = A$ and $\mathbf{w}(0) = \mathbf{c}(0) = 0$. Ψ is the effective pressure: gravity lowers it, and it turns negative (the wave is Jeans unstable) while $k^2 < 4\pi G\rho_0/c_s^2$, i.e. near $k_x = 0$ when $k_y^2 < 4\pi G\rho_0/c_s^2$. That window, together with the epicyclic and shear terms, gives the transient swing amplification (Goldreich & Lynden-Bell 1965; Julian & Toomre 1966; Toomre 1981; the magnetized geometry is that of Kim & Ostriker 2001). The system is linear, so the amplitudes scale with A, and the figures plot δ/A and c_y/A.

A.4 The discrete Laplacian

The multigrid solver inverts the 7-point Laplacian, whose eigenvalue for $e^{i\theta}$ on a grid of spacing $\Delta x = \Delta y$ is

$$D_h = -\left(\frac{2}{\Delta x}\right)^2\left[\sin^2\frac{k_x\Delta x}{2} + \sin^2\frac{k_y\Delta y}{2}\right] \tag{13}$$

($k_z = 0$ removes the third term), which tends to $-k^2$ as $k\Delta x \to 0$. The comparisons use D_h at the root spacing ($\Delta x = 2\pi/128$), the grid on which most of the wave lives in the refined runs. With $4\pi G\rho_0 = 1.9$, $k_{x0} = -6$, $k_y = 1$ the difference is small:

B_0	peak δ/A (time), continuum	the same, D_h at 128^2	$\max_{t\le 6}\lvert c_y\rvert/A$
0	-3.6090 (7.811)	-3.6083 (7.812)	—
0.5	$+2.2938$ (7.117)	$+2.2994$ (7.117)	1.566
1.0	-2.8454 (7.523)	-2.8469 (7.523)	1.997 / 1.998

(Peak over $t \le 8$.) The field reduces the amplification: magnetic pressure stiffens the compression (Ψ is effectively raised by v_A^2 for radial motions, see below), and the azimuthal field carries the oscillation past the swing (Fig. 5).

A.5 Checks

Divergence. $\nabla \cdot \mathbf{b} = -(k_x c_x + k_y c_y)\sin\theta$, and

$$\frac{d}{dt}(k_x c_x + k_y c_y) = q\Omega k_y c_x + k_x k_y B_0 w_x + k_y(-k_x B_0 w_x - q\Omega c_x) = 0,$$

so a divergence-free initial field stays divergence-free; the integrator keeps $k_x c_x + k_y c_y$ at 10^{-11} of A.

Axisymmetric limit. For $k_y = 0$, k_x is constant, c_x stays zero, and the system closes on w_x:

$$\ddot{w}_x = -\left[\kappa^2 + k_x^2\left(c_s^2 + v_A^2\right) - 4\pi G\rho_0\right] w_x,$$

the dispersion relation of axisymmetric magnetosonic–epicyclic waves in a self-gravitating slab, $\omega^2 = \kappa^2 + k_x^2(c_s^2 + v_A^2) - 4\pi G\rho_0$ (with $D = -k_x^2$): the field adds its pressure to the gas pressure for compressions perpendicular to it.

Hydrodynamic limit. For $B_0 = 0$ the field decouples ($\mathbf{c}$ is driven only through B_0) and the first three equations are the self-gravitating hydrodynamic swing of §7. Its peak is -3.609 at $t = 7.811$; at $t = 7.804$, the output time at which the runs peak, it is -3.605, the value quoted there.

A.6 Projections and the comparison

The history output (`swing.cpp`) projects the solution on the instantaneous wave,

$$\texttt{d_cos} = \frac{2}{V}\int\left(\frac{\rho}{\rho_0} - 1\right)\cos\theta\, dV, \quad \texttt{vx_sin} = \frac{2}{V}\int v_x \sin\theta\, dV, \quad \texttt{by_cos} = \frac{2}{V}\int \delta B_y \cos\theta\, dV,$$

and likewise `vy_sin`, `bx_cos`, with θ built from $k_x(t)$. For a single linear wave these return δ, w_x, w_y, c_x, c_y exactly; the sine projection of the density, `d_sin`, should stay zero, and its size measures phase errors. Without orbital advection v_y includes the shear flow, whose projection vanishes ($\int x\sin\theta\, dV = 0$ over a box with $k_y \neq 0$), so the same projections serve both frames. The tables report, for a variable f (here δ or c_y), the trajectory deviation $\max_t |f - f_{\rm ref}| / \max_t |f_{\rm ref}|$ over the stated interval, against linear theory (amplitude-scaled) or against the uniform twin interpolated to common times. The integration uses an eighth-order Runge–Kutta method (DOP853) with relative tolerance 10^{-11}, so the theory curves are exact to plotting precision.